

Change of Coordinate Matrix in $\mathbb{R}^n$

We call the “standard basis” $\mathcal{S} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and assume we have $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ as a basis of our vector space V , then we defined

$$P(\mathcal{B})_{\mathcal{S}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{S}} & \dots & [\mathbf{b}_n]_{\mathcal{S}} \end{bmatrix}.$$

The above represents our basis vectors as an array of real numbers.

What does this mean for $V = \mathbb{R}^n$? Let's compute $P(\mathcal{B})_{\mathcal{S}}\mathbf{x}$!

$$P(\mathcal{B})_{\mathcal{S}}\mathbf{x} = x_1 [\mathbf{b}_1]_{\mathcal{S}} + \dots + x_n [\mathbf{b}_n]_{\mathcal{S}}$$

This means if $\mathbf{x}$ is an array of coordinates for the $\mathcal{B}$ basis, then the output is a vector written in the standard basis!

Change of Coordinate Matrix in $\mathbb{R}^n$ Example

Consider the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Then, $P(\mathcal{B})_{\mathcal{S}} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$. So, if we say that $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$, and we want to find what it is in the standard basis we compute

$$\mathbf{x} = P(\mathcal{B})_{\mathcal{S}} [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + 4 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \\ 4 \end{bmatrix}$$

Change of Coordinate Matrix in $\mathbb{R}^n$ is a Linear Transformation

Theorem

If $V = \mathbb{R}^n$, then the Coordinate Matrix $P(\mathcal{B})_{\mathcal{S}}$ is a Linear Transformation given by

$$P(\mathcal{B})_{\mathcal{S}} : [\mathbf{x}]_{\mathcal{B}} \mapsto \mathbf{x},$$

which changes a vector from $\mathcal{B}$ coordinates to being written in the standard basis and

$$P(\mathcal{B})_{\mathcal{S}}^{-1} : \mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$$

changes a vector from the standard basis back to $\mathcal{B}$ coordinates.

Changing Coordinate From One Basis to Another

In some data science applications, we may want to change problems from one basis to another.

Theorem

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases for a vector space V . Then, there exists a unique matrix $P_{\mathcal{C} \leftarrow \mathcal{B}} = P(\mathcal{C})_S^{-1} P(\mathcal{B})_S$

This is really just saying that we (1) change our vector from being written in $\mathcal{B}$ coordinates to being written in the standard basis, then (2) change our vector from the standard basis to $\mathcal{C}$ coordinates.

Is there a potentially better way?

A Better Way Without Explicitly Computing an Inverse

Yes!

We also have that

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = P(\mathcal{B})_{\mathcal{C}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}$$

Which we can solve in one of two ways

1) Solve

$$\begin{bmatrix} P(\mathcal{C})_{\mathcal{S}} & | & P(\mathcal{B})_{\mathcal{S}} \end{bmatrix}$$

Which is solving the equation

$$P(\mathcal{C})_{\mathcal{S}} X = P(\mathcal{B})_{\mathcal{S}}$$

2) Solve

$$\begin{bmatrix} P(\mathcal{C})_{\mathcal{S}} & | & \mathbf{y} \end{bmatrix}$$

Which is solving the equation

$$P(\mathcal{C})_{\mathcal{S}} \mathbf{x} = \mathbf{y}$$

then plugging in $\mathbf{y} = \mathbf{b}_1, \dots, \mathbf{b}_n$.

Either will give the right answer, it's just a matter of preference!

Change of Coordinate Example

Consider the following bases of $\mathbb{R}^3$.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Compute $P_{\mathcal{C} \leftarrow \mathcal{B}}$. First, we try method 1.

$$\left[P(\mathcal{C})_{\mathcal{S}} \mid P(\mathcal{B})_{\mathcal{S}} \right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 2 & -2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 1 & -1 \end{array} \right]$$

So

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$$

Change of Coordinate Example

Consider the following bases of $\mathbb{R}^3$.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Compute $P_{\mathcal{C} \leftarrow \mathcal{B}}$. Next, we try method 2.

$$\left[P(\mathcal{C})_{\mathcal{S}} \mid \mathbf{y} \right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & y_1 \\ 1 & 1 & 0 & y_2 \\ 1 & 1 & 1 & y_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & y_1 \\ 0 & 1 & 0 & y_2 - y_1 \\ 0 & 1 & 1 & y_3 - y_1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & y_1 \\ 0 & 1 & 0 & y_2 - y_1 \\ 0 & 0 & 1 & y_3 - y_2 \end{array} \right]$$

So the columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are given by

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, [\mathbf{b}_3]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} \rightarrow P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$$

Change of Coordinate Practice

Consider the following bases of $\mathbb{R}^3$.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Compute $P_{\mathcal{C} \leftarrow \mathcal{B}}$

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{3}{4} \\ \frac{1}{2} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

Matrix Vector Space Practice

Let V be the vector space of **symmetric** 2×2 real matrices. We have the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

Consider the set $\mathcal{C} = \left\{ \begin{bmatrix} 1 & -2 \\ -2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \right\}$. Answer the following questions

1) Show that $\mathcal{C}$ is a basis for V .

Set up and show

$$P_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 2 & -1 \\ 2 & 4 & 1 \end{bmatrix}$$

is invertible.

2) Write the matrix corresponding to

$$\left[\begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \right]_{\mathcal{C}} \text{ in } \mathcal{B} \text{ coordinates}$$

$$\begin{bmatrix} -4 & 4 \\ 4 & 14 \end{bmatrix}$$